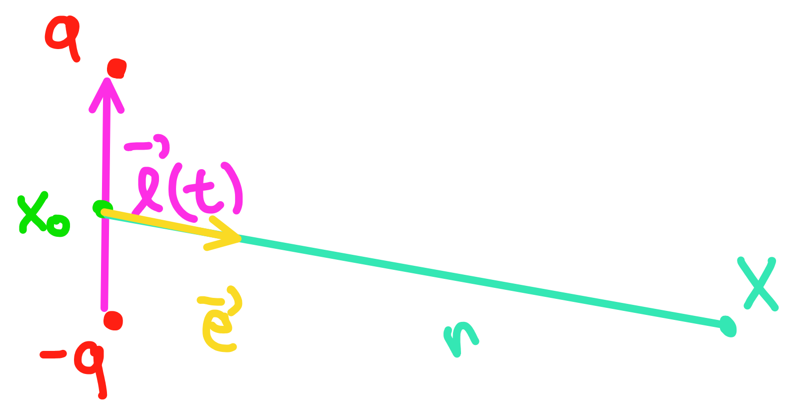
Dipólové záření



$$\vec{d}(t) = q\vec{\ell}(t)$$

$$\rho = -\vec{d}\cdot\vec{\nabla}\delta_{x_0}$$

$$\vec{j} = \dot{\vec{d}}\,\delta_{x_0}$$

$\big\downarrow\ x_0 \equiv 0$

$$\vec{A} = \frac{\mu}{4\pi}\int \frac{\vec{j}\left(t-\frac{r(x|x')}{c}\right)}{r(x|x')}\,dx' = \frac{\mu}{4\pi}\,\frac{\dot{\vec{d}}\left(t-\frac{r}{c}\right)}{r}$$

$$\phi = \frac{1}{4\pi\varepsilon}\int \frac{\rho\left(t-\frac{r(x|x')}{c}, x'\right)}{r(x|x')}\,dx' \overset{\text{přehodím}\ \vec{\nabla}'}{=}$$

$$= \frac{1}{4\pi\varepsilon}\int\left[\vec{\nabla}'\cdot\frac{\vec{d}\left(t-\frac{r(x|x')}{c}\right)}{r(x|x')}\right]\delta_0\,dx'$$

↑ ji vecpat (pozor!)

$\vec{\nabla}' r = -\vec{e}$

$$= \frac{1}{4\pi\varepsilon}\left[\frac{\dot{\vec{d}}\left(t-\frac{r}{c}\right)\left(-\frac{1}{c}\right)\left(-\vec{e}\right)}{r} - \frac{\vec{d}\left(t-\frac{r}{c}\right)}{r^2}\cdot\vec{e}\right]$$

$$= \frac{1}{4\pi\varepsilon}\left[\frac{\dot{\vec{d}}\left(t-\frac{r}{c}\right)\cdot\vec{e}}{rc} - \frac{\vec{d}\left(t-\frac{r}{c}\right)}{r^2}\cdot\vec{e}\right]$$

↓ Chceme $\vec{E}$ a $\vec{B}$, ale uvažujeme jen členy $\propto \frac{1}{r}$, jiné v dálce vymizí + $\vec{\nabla}\vec{e} \sim \frac{1}{r^2}$

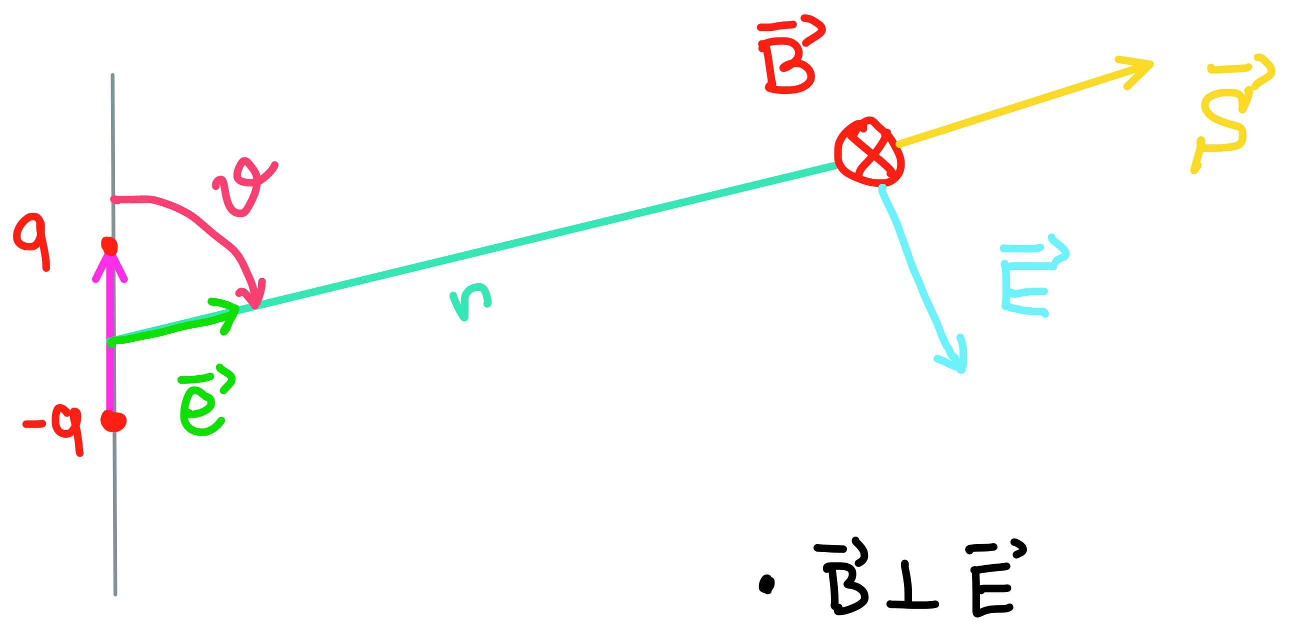
$$\vec{B} = \vec{\nabla}\times\vec{A} = -\frac{\mu}{4\pi}\,\frac{\vec{e}\times\ddot{\vec{d}}\left(t-\frac{r}{c}\right)}{cr} + O\left(\frac{1}{r^2}\right)$$

$$\vec{E} = -\vec{\nabla}\phi - \frac{\partial\vec{A}}{\partial t} = -\frac{1}{4\pi\varepsilon}\left(-\frac{\vec{e}\,\ddot{\vec{d}}\left(t-\frac{r}{c}\right)\cdot\vec{e}}{c^2 r}\right) - \overset{\frac{1}{4\pi\varepsilon c^2}}{\cancel{\frac{\mu}{4\pi}}}\,\frac{\ddot{\vec{d}}\left(t-\frac{r}{c}\right)}{r}$$

$$= +\frac{1}{4\pi\varepsilon c^2}\,\frac{\vec{e}\times\left[\vec{e}\times\ddot{\vec{d}}\left(t-\frac{r}{c}\right)\right]}{r} + O\left(\frac{1}{r^2}\right)$$

$$\vec{B} = -\frac{1}{4\pi\varepsilon c^2}\,\frac{\vec{e}\times\ddot{\vec{d}}\left(t-\frac{r}{c}\right)}{cr} + O\left(\frac{1}{r^2}\right)$$

$$\vec{E} = \frac{1}{4\pi\varepsilon c^2}\,\frac{\vec{e}\times\left[\vec{e}\times\ddot{\vec{d}}\left(t-\frac{r}{c}\right)\right]}{r} + O\left(\frac{1}{r^2}\right)$$



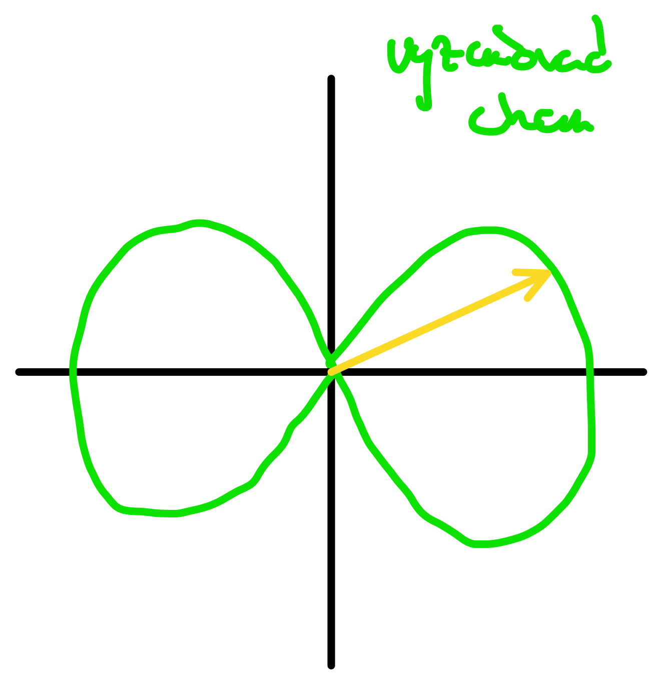
- $\vec{B}\perp\vec{E}$
- $E = cB$

$$\vec{S} = \vec{E}\times\vec{H} = \varepsilon c^2\,\vec{E}\times\vec{B} = \varepsilon_0 c^3 B^2\vec{e}$$

$$= \frac{1}{(4\pi)^2\varepsilon c^3}\,\frac{\left|\vec{e}\times\ddot{\vec{d}}_{ret}\right|^2}{r^2}\,\vec{e}$$

$$\vec{S} = \frac{1}{(4\pi)^2\varepsilon c^3}\,\frac{\ddot{d}_{ret}^2\sin^2\vartheta}{r^2}\,\vec{e}$$

vyzařovací chara.



→ rychlost vyzařování!

$$\mathcal{R} = \int_{\partial V}\vec{S}\cdot d\vec{S} = \frac{\ddot{d}_{ret}^2}{(4\pi)^2\varepsilon c^3}\underbrace{\int\frac{\sin^2\vartheta}{r^2}\,r^2\sin^2\vartheta\,d\vartheta\,d\varphi}_{2\pi\frac{4}{3}}$$

∂V … velká sféra

$$\mathcal{R} = \frac{2}{3}\,\frac{\ddot{d}_{ret}^2}{4\pi\varepsilon c^3}$$